



Just as each card-holder has only a small part of the message, which means nothing out of context, each neuron's message has a meaning that depends on what the other neurons are doing at the same time.

tern provided by a few hundred fans might spell out "Go, State." The message of neurons is a little like that. The analogy falters because the message in the cards means something only if someone else on the other side of the field sees the pattern and reads it, whereas there is nothing to "read" the message of the neurons, except for the neurons themselves. But the analogy works in some regards: Each neuron conveys a message that means nothing by itself but becomes part of an important message in context. Also, a mistake by one card holder (or one neuron) is not too costly; the overall message still gets through. But systematic mistakes by a whole group of individuals (neurons) could destroy or garble the message.

S U M M A R Y

- ★ **Neuron structure.** A neuron, or nerve cell, consists of a cell body, dendrites, and an axon. The axon conveys information to other neurons. (page 75)
- ★ **The action potential.** Information is conveyed along an axon by an action potential, which is regenerated without loss of strength at each point along the axon. (page 77)
- ★ **Mechanism of the action potential.** An action potential depends on the entry of sodium into the axon. Anything that blocks this flow will block the action potential. (page 77)
- ★ **How neurons communicate.** A neuron communicates with another neuron by releasing a chemical called a neurotransmitter at a specialized junction called a synapse. A neurotransmitter can either excite or inhibit the next neuron. (page 78)

- ★ **Neurotransmitters and behavioral disorders.** An excess or a deficit of a particular neurotransmitter can lead to abnormal behavior, such as that exhibited by people suffering from Parkinson's disease. (page 81)

Suggestion for Further Reading

Kalat, J. W. (1998). *Biological psychology* (6th ed.). Belmont, CA: Wadsworth. Chapters 1 through 4 deal with the material discussed in this chapter in more detail.

Terms

- neuron** a cell of the nervous system that receives information and transmits it to other cells by conducting electrochemical impulses (page 75)
- glia** a cell of the nervous system that insulates neurons, removes waste materials (such as dead cells), and performs other supportive functions (page 75)
- cell body** the part of the neuron that contains the nucleus of the cell (page 75)
- dendrite** one of the widely branching structures of a neuron that receive transmissions from other neurons (page 75)
- axon** a single, long, thin, straight fiber that transmits information from a neuron to other neurons or to muscle cells (page 75)
- action potential** an excitation that travels along an axon at a constant strength, no matter how far it must travel (page 77)
- resting potential** electrical polarization that ordinarily occurs across the membrane of an axon that is not undergoing an action potential (page 77)

synapse the specialized junction between one neuron and another; at this point, one neuron releases a neurotransmitter, which either excites or inhibits the next neuron (page 78)

terminal bouton (or bouton) a bulge at the end of an axon from which the axon releases a chemical called a neurotransmitter (page 78)

neurotransmitter a chemical that is stored in the terminal of an axon and that, when released, activates receptors of other neurons (page 78)

postsynaptic neuron a neuron on the receiving end of a synapse (page 78)

dopamine a neurotransmitter that promotes activity levels and facilitates movement (page 81)

Answers to Concept Checks

6. You will not feel the pain immediately because the action potential must travel from your foot to your brain. (If the action potentials travel at, say, 30 meters per second and your toe is about 1.5 meters from your brain, you will feel the pain about 0.05 of a second after you stub your toe.) (page 78)
7. Increased branching of the axons and dendrites will increase the number of synapses. (page 80)
8. Under the influence of a drug that prevents norepinephrine from attaching to its receptors, the postsynaptic neuron will receive less inhibition than usual. If we presume that the neuron continues to receive a certain amount of excitation, it will then produce action potentials more frequently than usual. (page 80)
9. Haloperidol would increase the severity of Parkinson's disease. In fact, large doses of haloperidol can induce symptoms of Parkinson's disease in anyone. (page 82)

Web Resources

M.M.M. Brain Tour: Building Blocks

suhep.phy.syr.edu/courses/modules/MM/Biology/biology2.html
Good illustrations and very readable text lead you through this tour of neurons, neurotransmitters, and neural organization, produced by the Mind and Machine MODULE team at Syracuse University.

Explanations of Behavior at the Level of the Nervous system

Does a person who loses part of the brain also lose part of the mind?

Why should psychologists care about the organization of the brain and the effects of brain damage? They have both practical and theoretical reasons.

The practical reason is that they need to distinguish between people who act strangely because they have had bad experiences and people who act strangely because they have a brain disorder. To do so, psychologists need to know how brain damage affects behavior.

The theoretical reason for studying the brain is simply that the study of brain damage helps to explain the organization of behavior. In some manner or another, behavior must be made up of component parts, but what are those parts? Is behavior composed of ideas? sensations? movements? personality characteristics? And how do the various components combine to produce the overall behavior and experience? One way to answer such questions is to examine the effects of brain damage.

A Survey of the Nervous System

Psychologists and biologists distinguish between the central nervous system and the peripheral nervous system. The **central nervous system** consists of *the brain and the spinal cord*. The central nervous system communicates with the rest of the body by the **peripheral nervous system**, which is composed of *bundles of axons between the spinal cord and the rest of the body*. The *peripheral nerves that communicate with the skin and muscles* are collectively called the **somatic nervous system**. Those that control the heart, stomach, and other organs are called the **autonomic nervous system**. Figure 3.25 summarizes these major divisions of the nervous system.

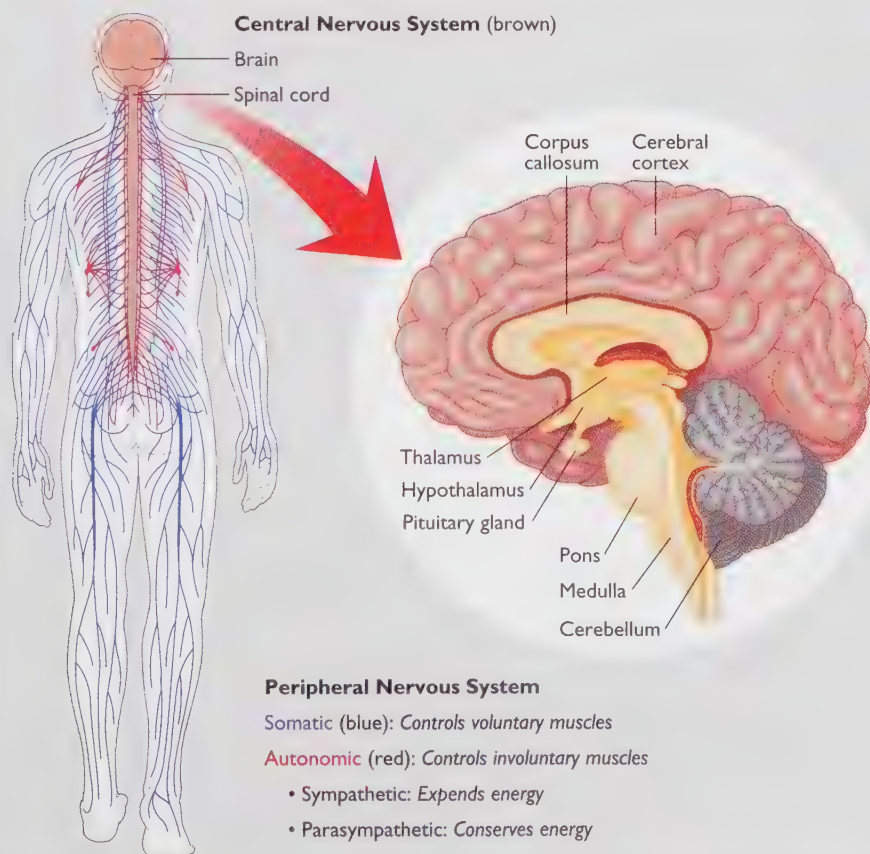


FIGURE 3.25 The nervous system has two major divisions; the central nervous system and the peripheral nervous system. Each of these has major subdivisions, as shown.

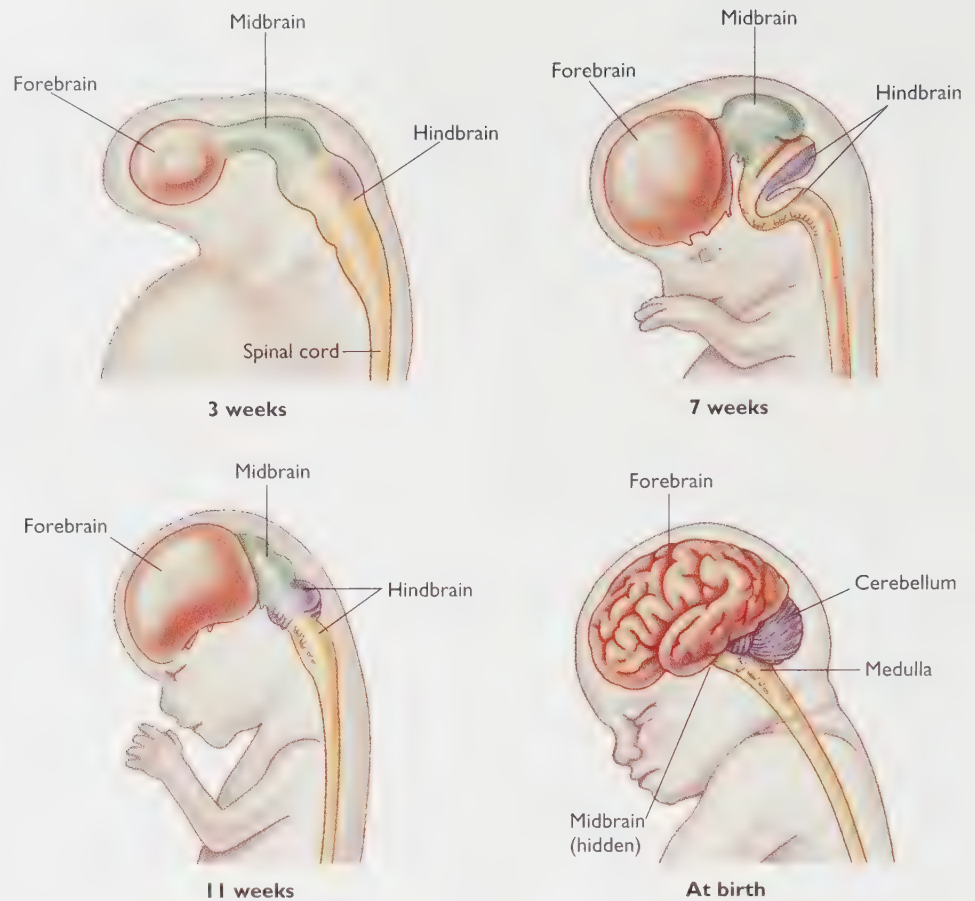


FIGURE 3.26 The human brain begins development as three lumps. By birth, the forebrain has grown much larger than either the midbrain or the hindbrain, although all three structures perform essential functions.

Early in its embryological development, the central nervous system of vertebrates, including humans, is a tube with three lumps, as shown in Figure 3.26. Those lumps develop into the *forebrain*, the *midbrain*, and the *hindbrain*; the rest of the tube develops into the spinal cord. The forebrain, which contains the cerebral cortex and other structures, is by far the dominant portion of the brain in mammals, especially in humans.

The Spinal Cord

The **spinal cord** communicates with the body below the level of the head by means of sensory neurons and motor neurons (Figure 3.27). The **sensory neurons** carry information about touch, pain, and other senses from the periphery of the body to the spinal cord. The **motor neurons** transmit impulses from the central nervous system to the muscles and glands.

The spinal cord serves both reflexive and voluntary behavior. A **reflex** is a *rapid, automatic response to a stimulus*. For example, suppose you put your hand on a hot stove. Stimulation of pain receptors in your finger sends messages along sensory neurons; within the spinal cord, these neurons send messages via interneurons to motor neurons, which then send impulses to the muscles that jerk your hand away from the hot stove.

The spinal cord is also necessary for voluntary behaviors. Every command to walk, talk, or make any other movement requires a message from the brain to the spinal cord and from the spinal cord to the muscles. People who have suffered damage to the spinal cord will lose some part of their muscle control. For example, after damage to the lower part of the spinal cord, a person cannot receive messages from the brain to the bottom of the spinal cord, which controls leg movements as well as bowel and bladder functions. After damage to the upper part of the spinal cord, a person would also lose control of the arm muscles.

The Autonomic Nervous System

The **autonomic nervous system**, closely associated with the spinal cord, controls the internal organs such as the heart. The term *autonomic* means involuntary or automatic. The autonomic nervous system is partly, though not entirely, automatic. We are generally unaware of its activity, although it does receive information from, and send information to, the brain and the spinal cord.

The autonomic nervous system consists of two parts: (1) *The sympathetic nervous system*, controlled by a chain of neurons lying just outside the spinal cord, increases heart rate and breathing rate and readies the body for vigorous fight-or-flight activities; and (2) *the parasympathetic nervous system*, con-

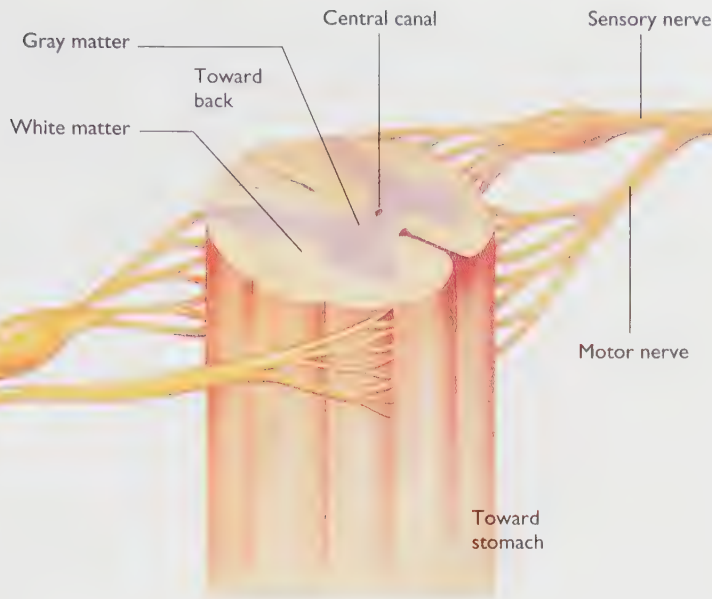


FIGURE 3.27 The spinal cord receives sensory information from all parts of the body except the head. Motor nerves in the spinal cord send messages to control the muscles and glands.

trolled by neurons at the very top and very bottom levels of the spinal cord, decreases heart rate, increases digestive activities, and in general promotes activities of the body that take place during rest (Figure 3.28). We shall return to this topic in more detail in the chapter about emotions (Chapter 12).

The Endocrine System

Although the endocrine system is not part of the nervous system, it is closely related to it—especially to the autonomic nervous system. Messages from the nervous system control the release of hormones from the endocrine system, and several hormones have major effects on certain kinds of brain activity.

The **endocrine system** is a set of glands that produce hormones and release them into the blood. Figure 3.29 shows some of the major endocrine glands. **Hormones** are chemicals released by glands and conveyed via the blood to other parts of the body, where they alter activity. Hormones are similar to neurotransmitters in that both affect the nervous system. The same chemical can be used both as

a hormone and as a neurotransmitter. The difference is that, when a chemical is used as a neurotransmitter, it is released immediately adjacent to the cell that it is to excite or inhibit. When it is used as a hormone, it is released into the blood, which diffuses it throughout the body.

Hormones can control effects that last from minutes (such as changes in blood pressure) to months (such as preparation for migration in birds). Some hormonal effects are nearly permanent. For example, the amount of the sex hormone testosterone present during prenatal development determines whether one develops a penis and scrotum or a clitoris and labia.

CONCEPT CHECK

- 10.** Just after a meal, the pancreas produces increased amounts of the hormone insulin, which increases the conversion of the digested food into fats in many cells throughout the body. How is a hormone more effective for this purpose than a neurotransmitter would be? (Check your answer on page 101.)

Sympathetic system

uses much energy

- Pupils open
- Saliva decreases
- Pulse quickens
- Sweat increases
- Stomach less active
- Epinephrine (adrenaline) secreted

Parasympathetic system

conserves energy

- Pupils constrict
- Saliva flows
- Pulse slows
- Sweat decreases
- Stomach churns

FIGURE 3.28 The sympathetic nervous system prepares the body for brief bouts of vigorous activity; the parasympathetic nervous system promotes digestion and other nonemergency functions. Although both systems are active at all times, the balance can shift from a predominance of one to a predominance of the other.

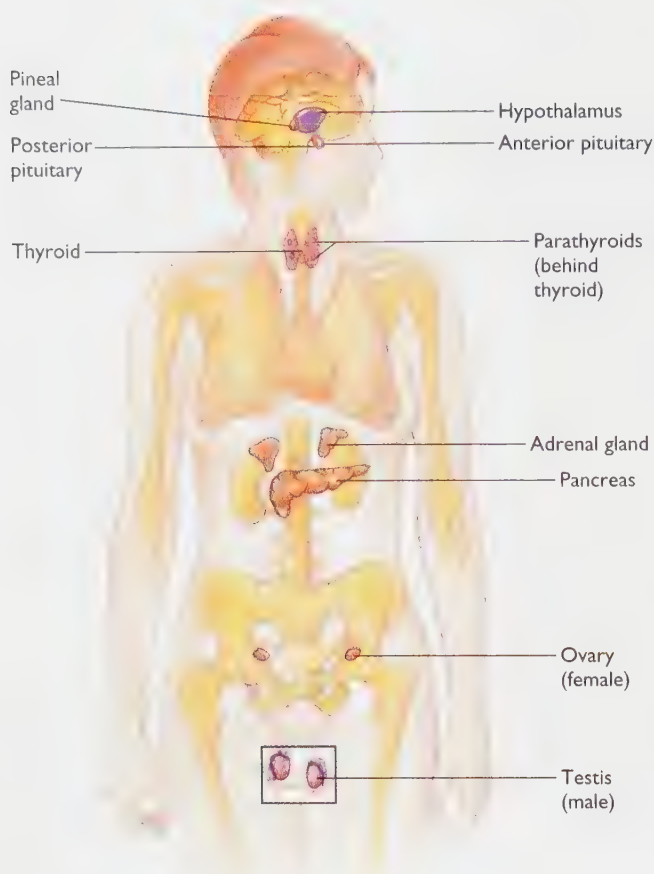


FIGURE 3.29 Glands in the endocrine system produce hormones and release them into the bloodstream.

Between the Spinal Cord and the Cerebral Cortex

Above the spinal cord, we find the hindbrain, the midbrain, the subcortical areas of the forebrain, and finally the cerebral cortex. (See Figure 3.30.) Information passing between the spinal cord and the cerebral cortex (in either direction) travels through other structures on the way. In some cases, that route is simple. For example, the primary motor area of the cerebral cortex sends axons that simply pass through other structures until they reach the spinal cord, where they make synapses onto cells that control the muscles. In other cases, the route is complex. For example, when you feel something with your fingers, the axons from your touch receptors send information to the spinal cord, which relays it to cells in the medulla (part of the hindbrain), which relay it to cells in the midbrain, which in turn relay it to cells in the thalamus (part of the forebrain), which in turn relay it to an area in the cerebral cortex. When I mention “relaying” information, I do not mean that the cells literally take the information and pass it along. At each stage, the cells combine and contrast this information with other information, processing it and changing it in various ways.

Furthermore, structures along the way have functions of their own. The **medulla** and **pons**, structures in the *hindbrain* (Figure 3.30), receive sensory input from the head (taste, hearing, touch sensations on the scalp) and send impulses for motor control of the head (for example, chewing, swallowing, and breathing). They also have axons that control breathing, heart rate, and other life-preserving functions.

The medulla, pons, and midbrain also contain the *reticular formation* and several other systems that regulate overall arousal in the brain (Robbins & Everitt, 1995; Szymusiak, 1995). These systems send messages diffusely to most of the forebrain. A malfunction in one of these systems, depending on its nature and location, will render a person either persistently sleepy or persistently aroused.

The **cerebellum** (Latin for “little brain”), *another part of the hindbrain*, is active in the control of rapid sequences of actions, such as playing the piano or dribbling a basketball. A person who suffers damage to the cerebellum can still make muscular movements but must plan each series of movements slowly, one at a time, instead of executing them in a smooth sequence. The cerebellum is also important for any behavior that requires timing, such as tapping out a rhythm, judging which of two visual stimuli is moving faster, and judging whether the delay between one pair of sounds is shorter or longer than the delay between another pair (Ivry & Diener, 1991; Keele & Ivry, 1990).

In later chapters, we shall return to examine some of the other brain structures that you see in Figure 3.30. The hippocampus is important for storing memories; people with damage to the hippocampus can remember most of what happened before the damage but little of what happened afterward (Chapter 7). The hypothalamus makes critical contributions to eating, drinking, sexual behavior, and other emotional or motivated behaviors (Chapter 11).

CONCEPT CHECK

11. People who have become intoxicated on alcohol have slow, slurred speech and cannot walk a straight line. From these observations, which part of the brain would you guess that the alcohol has most greatly impaired? (Check your answer on page 101.)

The Organization and Functioning of the Brain

Somehow, the brain is responsible for all our experiences and all our actions. When we try to explain how this process happens, we immediately encounter a problem: We each experience ourselves as a unity. That is, there seems to be just one “I” who sees, hears, touches, thinks, decides, and acts. Nevertheless, the brain is composed of many separate

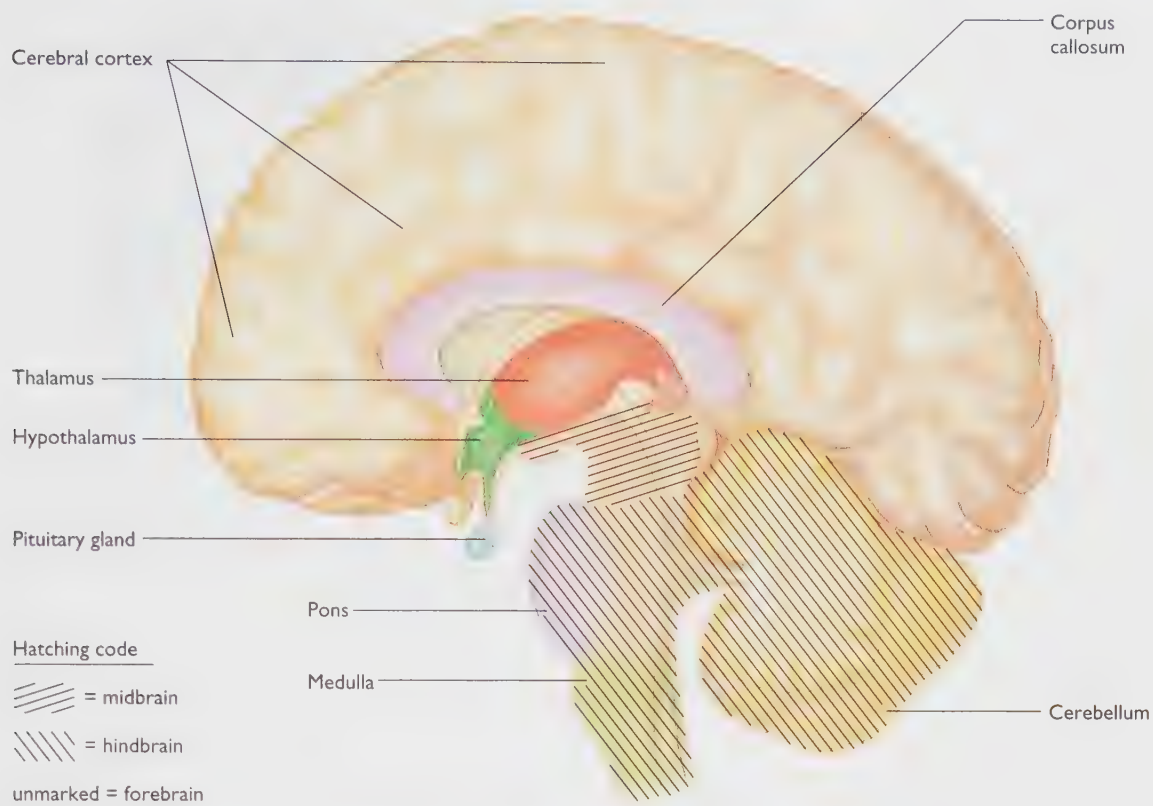


FIGURE 3.30 Top: The major divisions of the human central nervous system, as seen from the midline. Bottom: A side view of the brain, showing internal structures as though the cerebral cortex were transparent.

structures, each composed of substructures; overall, the brain includes tens of billions of neurons. How does one “I” emerge from all these separate voices?

A naive answer would be that all the various parts of the brain funnel their information to a “little person in the head”

who puts it all together. Although no one takes that concept seriously, the underlying idea is hard to abandon, and brain researchers have sometimes imagined a “master area” of the brain that would serve the same purpose as the little person—integrating all the information and making decisions.

That master area, if there is such a thing, would presumably be part of the cerebral cortex. The **cerebral cortex** is *the outer surface of the forebrain*—in humans, the largest and most prominent brain area. The cerebral cortex consists of two **hemispheres**, *the left and the right halves* (Figure 3.31). Each hemisphere is responsible for sensation and motor control on the opposite side of the body. (Why the reversal? Why does each hemisphere control the opposite side instead of its own side? Frankly, no one knows.) You have probably heard people talk about “having a lot of gray matter.” The cerebral cortex in the forebrain is *gray matter*; it contains a great many cell bodies, which are grayer than the axons. The interior of the forebrain beneath the cerebral cortex contains large numbers of axons, many of them covered with *myelin*, a white insulation. You can see areas of gray matter and white matter in Figure 3.32.

Research on the cerebral cortex, however, has found no “master area” or “central processor.” Researchers find few neurons that receive a combination of visual and auditory information or visual and touch information. Apparently, the sensory information does *not* all funnel into one central processor.

Then how does the cortex work? Each part of the brain has some specialized functions but also contributes to a variety of others. By analogy, a medical doctor has a specialized job but also performs many other general functions. If the doctor suddenly left town, we would mainly notice the loss at the hospital but also the losses in the doctor’s contributions to home maintenance, feeding

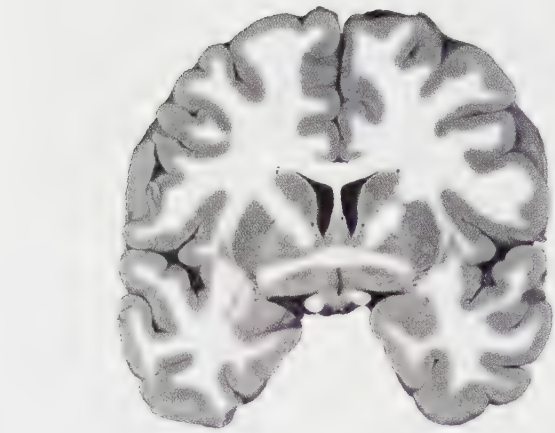


FIGURE 3.32 This cross section of a human brain shows the distinction between gray matter (composed mostly of cell bodies and dendrites) and white matter (composed almost entirely of axons). Myelin, a fatty sheath that surrounds many axons, makes the white matter white.

the family, chauffeuring the children, paying bills, community volunteer activities, and so forth. If the doctor was gone for long, other people would have to take over some of the uncompleted tasks, and as a result these people would become less efficient at their own specialized jobs—and so forth. Similarly, when a brain area becomes damaged or impaired, we primarily notice the loss of its specialized functions, but a variety of other functions will suffer a bit, too, and as other brain areas compensate for the loss, they may become inefficient at their own specialized tasks.

Still, the effects of brain damage sometimes include remarkably limited behavioral deficits. Let us examine some examples of brain damage and the specialized deficits that result.

Effects of Localized Brain Damage: The Visual System

The visual, auditory, and other systems of the brain largely operate independently of one another. The mystery deepens: Even within a given system, such as vision, the brain processes different aspects of the stimulus separately and in parallel. Figure 3.33 illustrates three major pathways in the visual areas of the human cerebral cortex. The main purpose of this figure is to illustrate the idea of separate pathways, not their exact locations. All three pathways begin in the *primary visual cortex*, designated here as V1 (meaning visual area 1), although they each begin with different neurons within that area. These pathways send their information to different neurons in the *secondary visual area*, V2, and from there the information branches out to additional areas. One pathway is mostly responsible for analysis of shape details; another, for color and brightness; another, for motion.



FIGURE 3.31 The human cerebral cortex: (bottom) left and right hemispheres; (top) inside view of a complete hemisphere. The folds greatly extend the brain’s surface area.

A person who suffers damage to the entire primary visual cortex (V1) will become completely blind. In contrast, consider what happens to someone with damage to only part of a single pathway. After damage to parts of the inferior temporal cortex, which is part of the shape pathway, some people suffer a specific **faceblindness**, a *specific inability to recognize people by their faces*. Such a person may still be able to read and can see faces well enough to say, “That person is almost bald, has a somewhat thin face, and dark eyes,” but lacks the ability to say, “Aha, that’s my cousin Ross.” The affected person may also fail to identify specific plants, animals, cars, and so forth (Farah, 1990; 1992).

In contrast, people with damage to parts of the color pathway lose an ability called **color constancy**, which is *the ability to continue recognizing colors even after a change in lighting*. For example, suppose you put on blue-tinted glasses. You would not suddenly see everything as blue; you could still identify which objects are red, yellow, and so forth. People with certain kinds of brain damage have trouble with color constancy; for these people, everything in the room would indeed look blue, viewed through the glasses (Zeki, 1980; 1983).

Damage in the motion pathway produces a very surprising deficit: **motionblindness**. People with such damage have *trouble identifying the movement of objects* (Zihl, von Cramon, & Mai, 1983). One such patient found

that she could not safely cross a road, because she could not see which cars were moving or how fast. She had trouble pouring coffee into a cup, because she could not see the liquid level gradually rising in the cup. (Eventually, she would stop when she saw coffee all over the table.) For most of us, it is difficult to imagine what the world must look like to a motionblind person. It might be a little bit like living in a world with only strobe lights shining once every few seconds, except that these people experience no blackouts in their perception.

In short, different parts of the visual system of your cerebral cortex process different aspects of each visual stimulus—one part attends to shape, another to color, another to motion. How does your brain put it all together? Again, we have reached the limitations of current knowledge; although the brain must somehow combine different kinds of information, researchers do not currently understand how it does this.

Effects of Localized Brain Damage: Unilateral Neglect

Let’s now consider a kind of brain damage that affects attention rather than sensation. Damage in the parietal lobe (see Figure 3.34), especially in the right parietal lobe, leads to **neglect** (*disregard*) of the opposite side of the body and, indeed, the opposite side of their world. Such people may

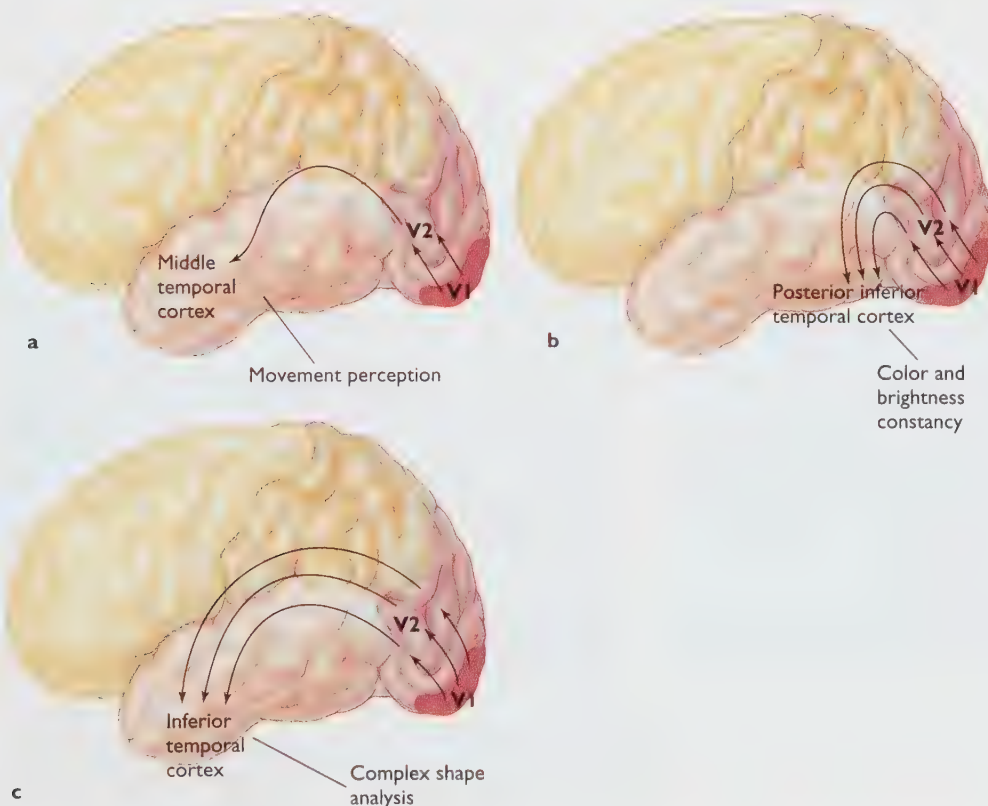


FIGURE 3.33 Three major visual pathways in the cerebral cortex: (a) One path is responsible for movement perception; (b) a second path deals with color and brightness; (c) the third path provides analysis of complex shapes. Because the three paths are largely independent of one another, brain damage can impair one aspect of visual perception without blocking the others.

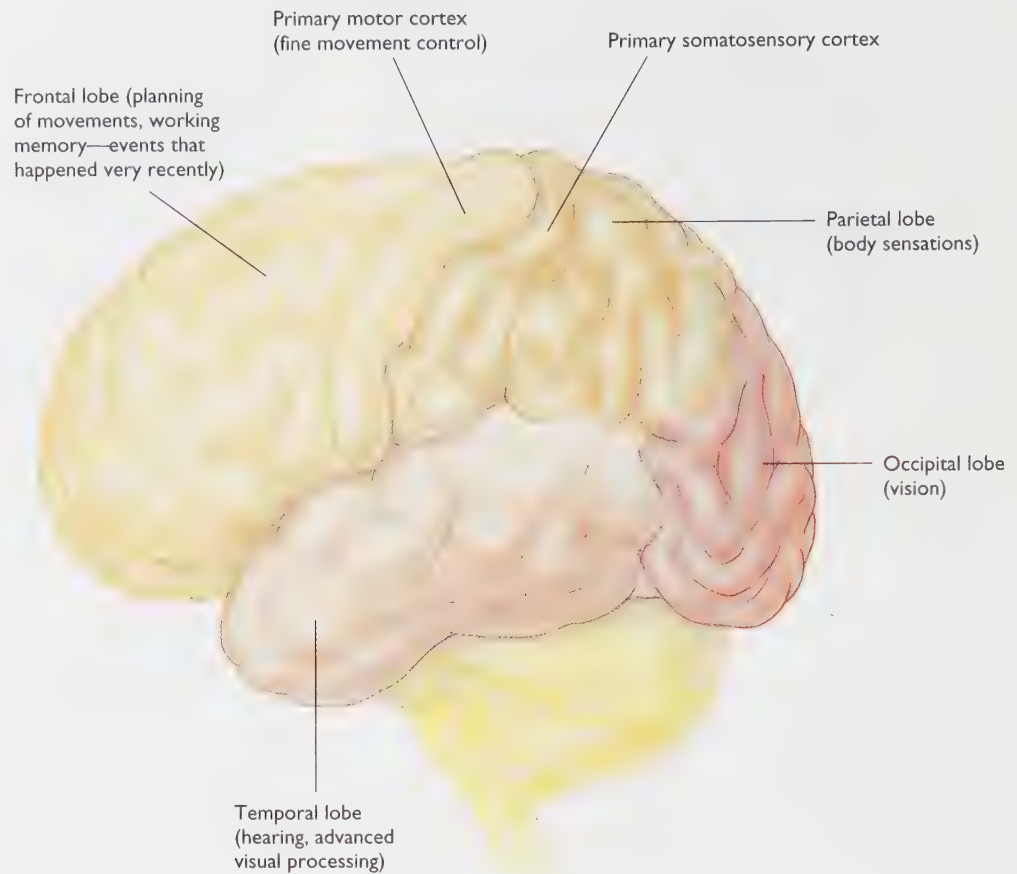


FIGURE 3.34 The four lobes of the human cerebral cortex, with indications of some of their major functions.

put clothing on only the right side of the body, insisting that the left half belongs to someone else. They read only the right side of a page and draw only the right side of an object, using only the right side of the paper (Heilman, 1979; see Figure 3.35).

If you were to tickle such a person simultaneously on the left and right sides of the wrist (of either hand, left or right), he or she would report the sensation on the right side of that wrist, ignoring the sensation on the left side. Now, suppose you turn the person's hand upside down and repeat the experiment. He or she now reports the sensation on what has *now* become the right side of the hand, as illustrated in Figure 3.36 (Moscovitch & Behrmann, 1994). That is, the person does not simply neglect a certain area of skin, but detects what part of the arm is on the left, and then neglects it. The impairment is not a loss of sensation; it is a loss of attention.

Effects of Localized Brain Damage: Specific Memory Loss

Brain-damaged people sometimes show remarkably specialized kinds of memory loss, which we shall discuss in Chapter 7. Here, I wish to describe an even more specific kind of memory loss, which has so far been demonstrated only in monkeys.

This brain damage occurs in a tiny portion of the frontal cortex, and the behavior to be measured is a kind of delayed response. **Delayed response** is *any task in which an individual gets a signal and then must wait before it can make the necessary response*. In one set of studies, monkeys were trained to stare at a dot on a screen, as shown in Figure 3.37. A signal light flashed briefly at some spot away from the fixation point. The monkey had to continue looking at the fixation point, until the light went off several seconds later, and then move its eyes to the location where the signal light had been. Monkeys with extensive damage to one area in the frontal lobes of the cerebral cortex showed severe deficits on the task, apparently forgetting where the signal was. With very tiny damage within this brain area, monkeys lose their spatial memory *for just one spot*. For example, a monkey might be unable to remember

FIGURE 3.35 A person with damage to the right parietal lobe will draw only the right side of an object, as this attempt to copy a picture of a flower shows. (From Heilman, 1979.)





FIGURE 3.36 People who have undergone damage to the parietal lobe of the right hemisphere of the brain neglect (ignore) stimuli on the left side of either wrist. After the hand is rotated to a new position, the patient ignores what is now on the left, even though he or she attended to the same skin in the previous position. That is, the problem is one of attention, not of simple sensation.

a spot directly at the top of the circle or a spot at the left (Goldman-Rakic, 1994). We know the monkey can still see these spots, because the monkey can respond to them on a task requiring no delay. But as soon as we introduce a short delay, the monkey's performance deteriorates. Evidently, the memory for just one location has been lost.

Conclusions About Brain Organization

We have considered examples of damage to the visual system, neglect of half of the body, and loss of a tiny area of spatial memory. I hope that you will find these examples interesting for their own sake, but beyond that, the main point is that the brain is full of separate systems with separate tasks. A small area of brain damage can lead to a signif-

icant loss of a narrowly defined behavioral function, with little loss of other functions.

Here's another way of stating this conclusion, with greater emphasis on the philosophical implications: *If part of the brain is lost, part of behavior and experience is lost as well. As far as we can tell, brain activity and "mind" are inseparable; we cannot have one without the other.*

Division of Labor: The Four Lobes of the Cerebral Cortex

In the process of telling you about certain kinds of brain damage, I have already presented the functions of certain parts of the cerebral cortex. Let's now survey the major areas of the cortex more systematically.

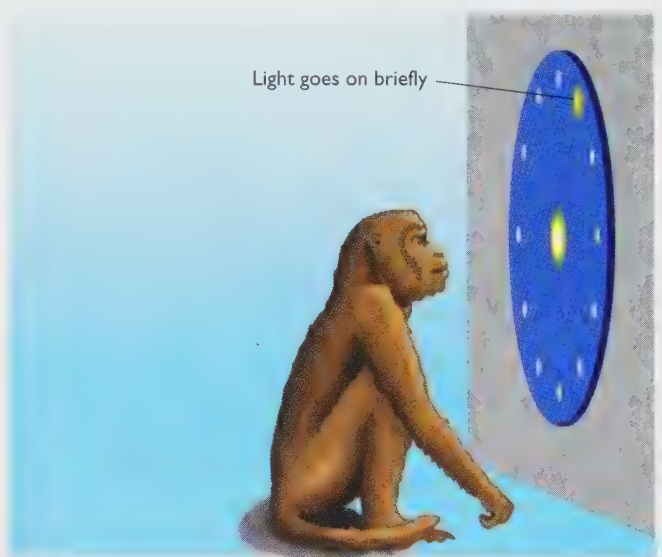
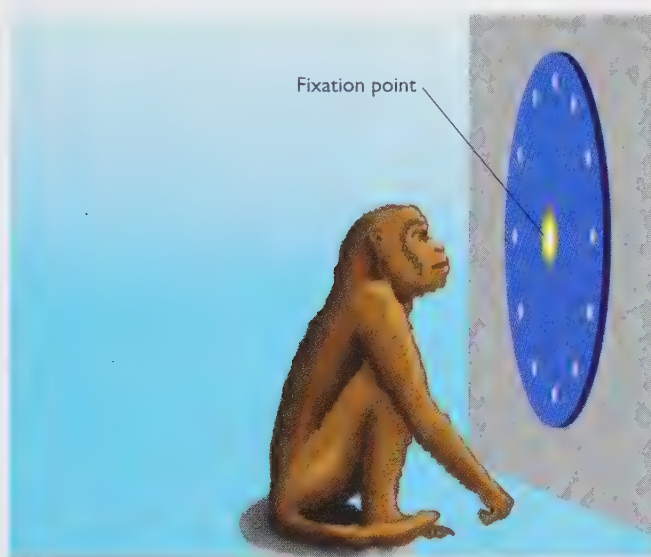


FIGURE 3.37 In one version of the delayed-response task, a monkey is trained to stare at a central fixation point. A light appears briefly at some point along the circle surrounding the fixation point. The monkey keeps its eyes focused on the fixation point during a delay and then must look at the point where the light appeared. The monkey must remember the location of the light during the delay.

For the sake of convenience, we describe the cerebral cortex in terms of four *lobes*—occipital, parietal, temporal, and frontal, as shown in Figure 3.34. The **occipital lobe**, *at the rear of the head, is specialized for vision*. As already discussed, different areas within (and outside) the occipital lobe contribute to different aspects of vision, such as shape, color, and motion. However, all of the occipital lobe apparently contributes to vision in one way or another.

The **parietal lobe**, *just anterior (forward) from the occipital lobe, is specialized for the body senses, including touch, pain, temperature, and awareness of the location of body parts*. The **primary somatosensory** (body-sensory) **cortex**, *a strip in the anterior portion of the parietal lobe, has neurons sensitive to touch in different body areas*, as shown in Figure 3.38. Note that in Figure 3.38a larger areas are devoted to touch in the more sensitive parts of the body,

such as the lips and hands, than to less sensitive areas, such as the abdomen and the back. Damage to any part of the somatosensory cortex will impair sensation from the corresponding part of the body.

The **temporal lobe** of each hemisphere, *located toward the left and right sides of the head, is the main processing area for hearing and also contributes to some of the more complex aspects of vision*. One area in the temporal lobe of the left hemisphere is important for language comprehension. Damage centered here impairs people's ability to understand what other people are saying; they also have trouble remembering the names of objects when they are speaking.

Other parts of the temporal lobe are critical for certain aspects of emotion. People with damage to one part of the temporal lobe, known as the *amygdala*, have almost

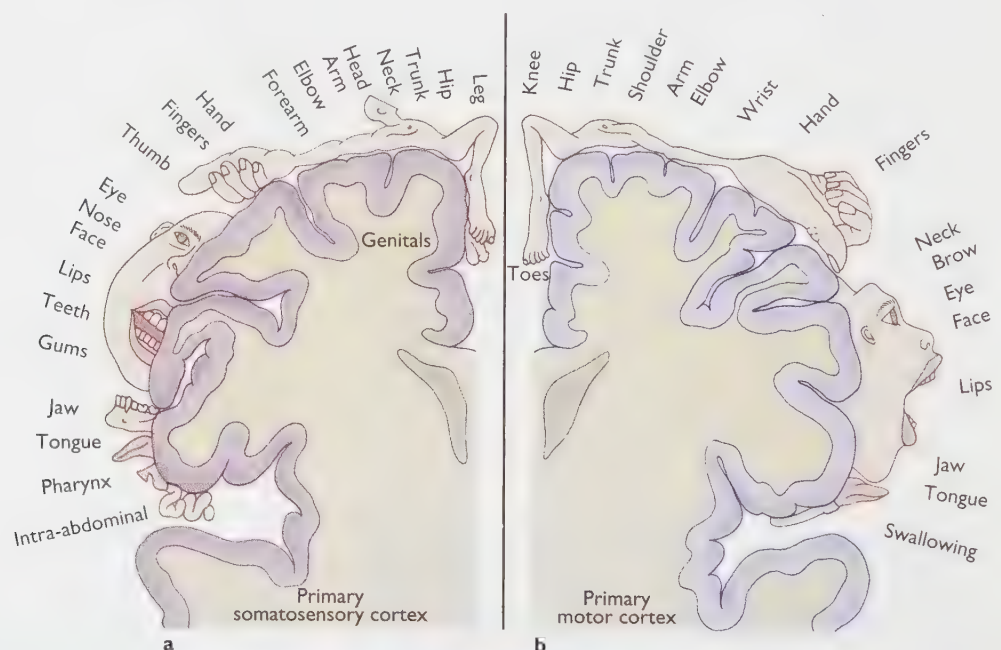
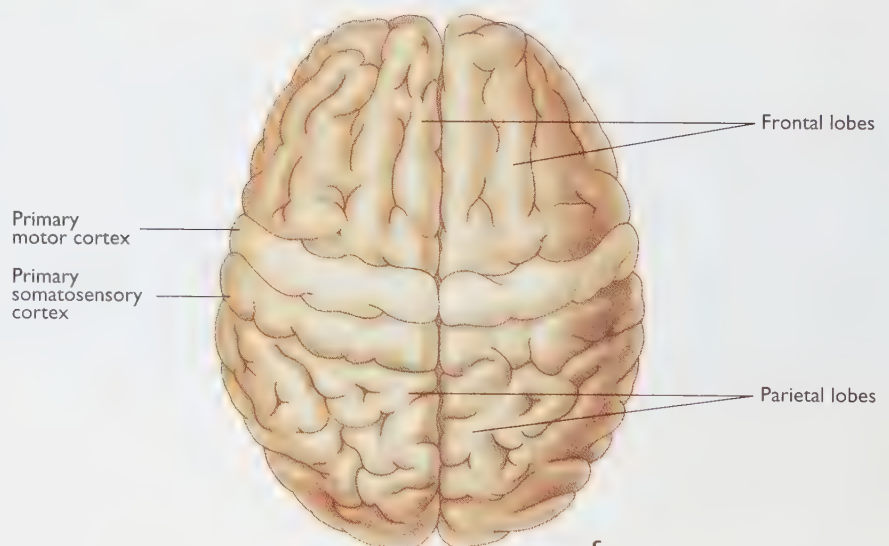


FIGURE 3.38 (a) The primary somatosensory cortex and (b) the primary motor cortex, illustrating which part of the body each brain area controls. Larger areas of the cortex are devoted to body parts that need to be controlled with great precision, such as the face and hands. The figure shows the left primary somatosensory cortex, which receives information from the right side of the body, and the right primary motor cortex, which controls the muscles on the left side of the body. (c) Locations of the primary somatosensory cortex and the primary motor cortex. [(a) and (b) from Geschwind, 1979.]



no fear or anxiety (LaBar, LeDoux, Spencer, & Phelps, 1995). Most people remember emotionally distressing events better than neutral events, but those with damage to the amygdala do not, presumably because they do not feel the emotional distress (Cahill, Babinsky, Markowitsch, & McGaugh, 1995). These people can neither recognize facial expressions of fear in others nor imagine what a frightened face would look like (Adolphs, Tranel, Damasio, & Damasio, 1994).

The **frontal lobe**, at the *anterior (forward) pole of the brain*, includes the **primary motor cortex** (Figure 3.38), a structure that is important for the control of fine movements, such as moving one finger at a time. As with the primary somatosensory cortex, each area of the primary motor cortex controls a different part of the body, and larger areas are devoted to precise movements of the tongue and fingers than to, say, the shoulder and elbow muscles. The *anterior sections of the frontal lobe*, called the **prefrontal cortex**, contribute to the organization and planning of movements and to certain aspects of memory. Indeed, planning a movement depends on memory. Recall, for example, the delayed response task (page 92): The individual must remember a signal during a delay and then make the appropriate movement.

Certain areas in the left frontal lobe are essential for human language production. People with extensive damage to the left frontal lobe have trouble speaking, writing, or gesturing in sign language (Bellugi, Poizner, & Klima, 1983; Geschwind, 1970). What they say still makes sense, although they generally omit prepositions, conjunctions, and word endings.

How do we know that these brain areas have the functions that I have described? For many years, nearly all the evidence came from observations of brain damage. Researchers can now supplement such evidence with modern techniques that measure activity in an unanesthetized brain (see Figure 3.39). For example, **functional magnetic resonance imaging (fMRI)** uses magnetic detectors outside the head to measure the amounts of hemoglobin, with and without oxygen, in different parts of the brain (Cohen, Noll, & Schneider, 1993). Brain areas that are highly active use much oxygen and therefore decrease the amount of oxygen bound to hemoglobin in the blood. The fMRI technique therefore provides a way of inferring which brain areas are currently more active than others. Figure 3.40 gives an example of an fMRI scan.

CONCEPT CHECK

- 12.** The following five people are known to have suffered damage to the cerebral cortex. From their behavioral symptoms, determine the probable location of the damage for each person: (a) impaired perception of the left half of the body and a tendency to ignore the left half of the body and the left half of the world; (b) impaired hearing and some changes in emotional experience; (c) inability to make fine movements with the right hand; (d) loss of vision in the left visual field; and (e) poor performance on a delayed response task, indicating difficulty remembering what has just happened. (Check your answers on page 101.)

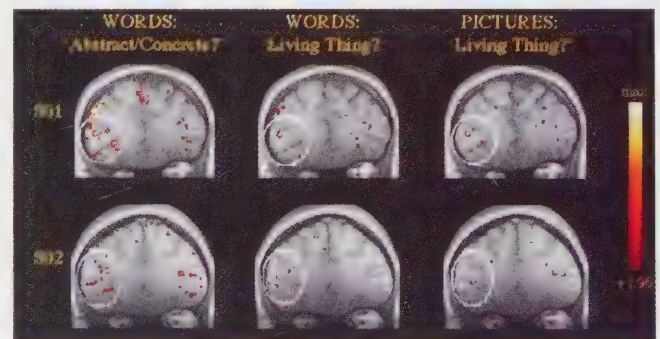
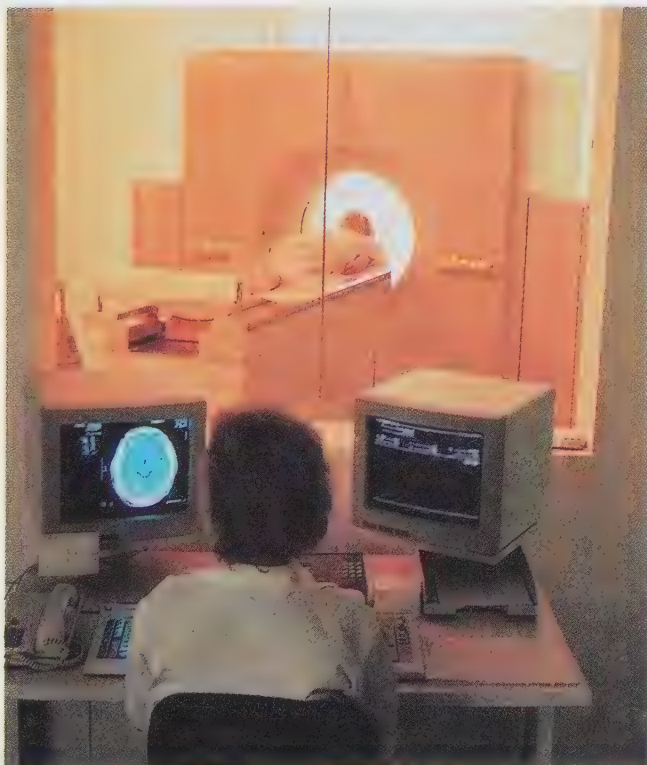


FIGURE 3.40 This brain scan was made with functional magnetic resonance imaging (fMRI). Participants looked at words or pictures and judged whether each item was abstract or concrete, living or nonliving. Yellow shows the areas most activated by this judgment; red shows areas less strongly activated. (From Wagner, Desmond, Domb, Glover, & Gabrieli, 1997. Photo courtesy of Anthony D. Wagner.)

FIGURE 3.39 Devices such as this computerized axial tomography (CAT) scanner produce detailed views of a living human brain.

Effects of Experience on Brain Structure

The drawing in Figure 3.25 shows “the human nervous system.” We do not need to specify whose nervous system, because the gross structure is about the same from one person to another. The detailed anatomy does vary, however, and some of these variations depend on people’s experiences.

Let’s begin with the effects of musical training. Figure 3.38(a) shows how the somatosensory cortex represents different parts of the body. One area, for example, represents the fingers, and researchers have measured the average amount of cortex devoted to the fingers. The results are different, however, for people who have played a stringed instrument (ordinarily, violin) since childhood. Right-handed stringed-instrument players hold the bow with the right hand and finger the strings with the left hand. The result of all those years of practice is that the amount of sensory cortex devoted to the left hand becomes about twice as large as usual (Elbert, Pantev, Wienbruch, Rockstroh, & Taub, 1995).

Many children have *absolute pitch* or “perfect pitch,” the ability to hear a note and identify it as B-flat, C-sharp, or whatever. Fewer than one person in 10,000 retains that ability into adulthood, and those few are people who began piano, voice, or other musical lessons in early childhood and continued practicing from then on (Takeuchi & Hulse, 1993). For those who maintain absolute pitch, one area of the temporal lobe of the cerebral cortex is significantly larger than normal (Schlaug, Jäncke, Huang, & Steinmetz, 1995).

What about people who are born blind and who learn to attend to touch stimuli more than other people do? One difference is that part of their occipital cortex, which for other people responds only to visual stimuli, becomes responsive to tactile stimuli, such as Braille characters (Cohen et al., 1997). The section of the somatosensory cortex that represents the index finger (the one ordinarily used for reading Braille) also becomes larger than usual. In fact, small changes in that area occur even from one day to the next; the representation of the index finger will be greater after a day of reading Braille than at the same time on a vacation day (Pascual-Leone, Wasserman, Sadato, & Hallett, 1995).

The biggest effects on brain anatomy appear in those who have had unusual and intensive experiences since early childhood, such as those who started musical training at age four or those who have been blind since birth. The immature brain is probably subject to greater modification than is the adult brain. However, to be fair, we should admit that the evidence does not fully separate the effects of the age of starting practice from the effects of total practice time. For example, a 24-year-old who started violin training at age four has had 20 years of practice, perhaps several hours per day. It would be hard to find, for comparison, a 44-year-old

who had started to learn the violin at age 24 and then worked at it for hours each day.

The general point is that our brain anatomy is not fixed at birth. Unusual or prolonged experiences can modify the structure of the brain to improve its efficiency at performing tasks that a given person commonly encounters.

The Corpus Callosum and the Split-Brain Phenomenon

What would happen if the sensory input from one half of your body, represented in one hemisphere, could not travel to the motor cortex in the opposite hemisphere? Such a situation does arise after damage to the **corpus callosum**, a set of axons connecting the two hemispheres (Figure 3.41). Corpus callosum damage also prevents someone from comparing sights seen on one side of their world against sights seen on the other side of the world and from comparing something felt with one hand to something felt with the other hand. Information that reaches one hemisphere stays in that hemisphere, because the person no longer has a single cerebral cortex, but rather has two half-cortexes operating side by side. Research on some unusual surgical patients suggests that the result of this operation is two separate spheres of consciousness.

Several teams of brain surgeons have cut the corpus callosum in an effort to relieve a condition called **epilepsy**, in which neurons somewhere in the brain begin to emit abnormal rhythmic, spontaneous impulses. Such impulses originate in different locations for different people and quickly spread to other areas of the brain, including neurons in the opposite hemisphere. The effects on behavior can vary widely, depending on where the epilepsy originates in the brain and where it spreads. Most people with epilepsy respond well to antiepileptic drugs and can live normal lives. A few people, however, do not respond to any of the known antiepileptic drugs and continue to have major seizures so frequently that they cannot work, go to school, or travel far from medical help. Such people are willing to try almost anything to get relief. In certain cases, surgeons recommended cutting the corpus callosum. The reasoning was that epileptic seizures would be prevented from spreading across the corpus callosum to the other hemisphere and would thus be less severe.

The operation was even more successful than expected. Not only were the seizures limited to one side of the body, but they also became far less frequent. A possible explanation is that the operation interrupted the feedback loop between the two hemispheres that allows an epileptic seizure to echo back and forth. These split-brain patients were able to return to work and to resume other normal activities. There were, however, some interesting behavioral side effects. But before I can discuss these, we need to consider some anatomy.



FIGURE 3.41 The corpus callosum is a large set of axons that convey information between the two hemispheres of the cerebral cortex. (left) A midline view showing the location of the corpus callosum. (right) A horizontal section showing how each axon of the corpus callosum links one spot in the left hemisphere to a corresponding spot in the right hemisphere.

Connections Between the Eyes and the Brain

Note: This section presents a simple concept that is contrary to most people's expectations. Even when students are warned that this material will appear on a test and are practically told what the question will be, many of them still answer incorrectly. So please pay attention!

Because each hemisphere of the brain controls the muscles on the opposite side of the body, each half of the brain needs to see the opposite side of the world. This does not mean that your left hemisphere sees with the right eye or that your right hemisphere sees with the left eye. Convince yourself: Close one eye, then open it and close the other. Note that you see almost the same view with both eyes. You see the left half of the world with part of your left eye and part of your right eye.

Figure 3.42, which shows the human visual system, warrants careful study. Light from each half of the world strikes receptors on the opposite side of *each* retina. Information from the left half of each retina travels via the *optic nerves* to the left hemisphere of the cerebral cortex; information from the right half of each retina travels via the optic nerves to the right hemisphere.

Here is one way to remember this material: *Light from each side of the world strikes the opposite side of the retina.*

The brain is connected to the eyes so that each hemisphere sees the opposite side of the world. If you remember those two statements, you should be able to deduce the connections shown in Figure 3.42.

What about the very center of the retina? The cells in a thin strip down the center of each retina send axons to both sides of the brain.

Behavioral Effects of Severing the Corpus Callosum

For almost all right-handed people and for about 60% of left-handed people, the brain area that controls speech is located in the left hemisphere of the brain. For most other left-handers, both hemispheres share control of speech to varying degrees. Complete right-hemisphere control of speech is possible, but rare. The right hemisphere does contribute to the emotional aspects of speech, however. People with right-hemisphere damage speak with little expression and have trouble interpreting the emotions that other people express through tone of voice (Shapiro & Danly, 1985; Tucker, 1981).

If you have the same brain organization as most other people, you can describe only the information that reaches your left hemisphere. However, any information that first enters your right hemisphere passes quickly across the

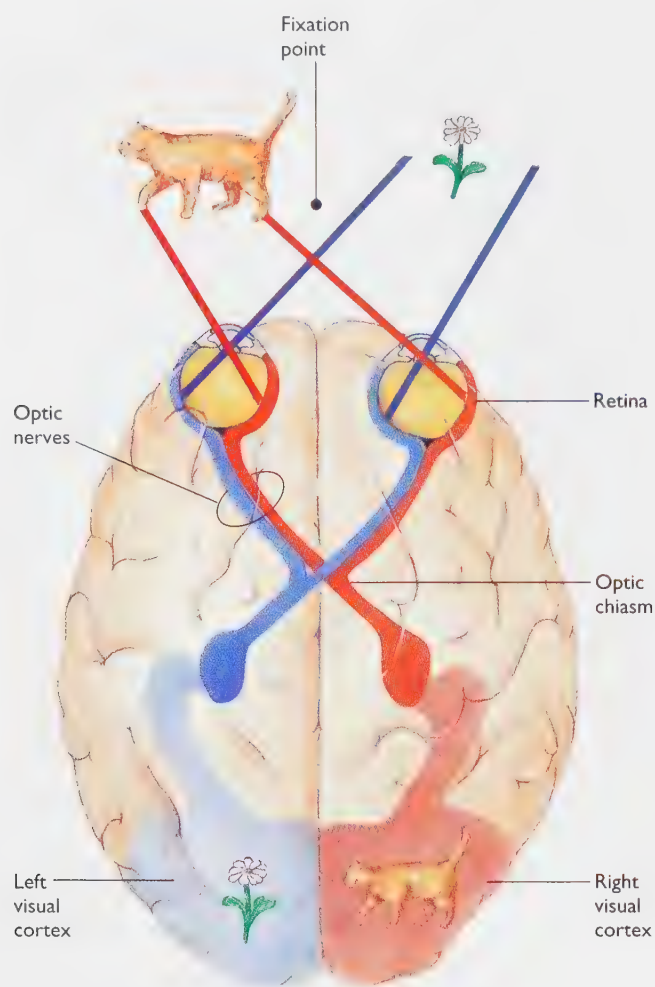


FIGURE 3.42 In the human visual system (viewed here from above), light from either half of the world crosses through the pupils to strike the opposite side of each retina. Axons from the left half of each retina travel to the left hemisphere of the brain; axons from the right half of each retina travel to the right hemisphere of the brain.

corpus callosum to your left hemisphere, so you can talk about anything that reaches any receptor in your body.

But what happens when the corpus callosum is severed? When a woman with a severed corpus callosum touches something with her right hand without looking at it, she can say what it is, because the touch information reaches her left hemisphere (Nebes, 1974; Sperry, 1967). However, if she feels something with her left hand, then she cannot say what it is, because the information reaches only her right hemisphere. If she is given several choices and is asked to point to what her left hand has just felt, she can point to it correctly—but only with her left hand. In fact, she will sometimes point to the correct object with her left hand while saying, “I have no idea what it was. I didn’t feel anything.” Evidently, the right hemisphere can understand the instructions and answer with the hand it controls, but it cannot talk. Roger Sperry won a Nobel Prize in physiology and medicine in 1981 for these pioneering discoveries.

Now consider what happens when this split-brain woman looks at something (Figure 3.43). Under ordinary conditions, when her eyes are free to move about, she sees almost the same thing in both hemispheres. In the laboratory, however, it is possible to restrict information to one side of the brain or the other by presenting the information faster than the eyes can move. The woman in Figure 3.43 focuses her eyes on a point in the middle of the screen. The investigator flashes a word such as *hatband* on the screen for only a split second so that the woman does not have enough time to move her eyes. If she is asked what she saw, she replies, “band,” which is what the left hemisphere saw. (Information from the right side of the screen, you will recall, goes to the left side of each retina and from there to the left hemisphere of the brain.) If she is asked what *kind* of band it might be, she will be puzzled: “I don’t know. jazz band? rubber band?” However, if she is asked to point with the left hand at what she saw, the left hand points to a hat (what the right hemisphere saw).

Split-brain people get along reasonably well in everyday life. Walking, for example, is no problem; it is largely controlled by subcortical areas of the brain that exchange information through connections below the corpus callosum.

In special circumstances, the two hemispheres find clever ways to cooperate. In one experiment, a split-brain person was looking at pictures flashed on a screen, as in Figure 3.43a. He could not name most of the objects flashed in the left visual field, but after some delay, he could name such simple shapes as round, square, or triangular. Here is how he did it: After seeing the object (with the right hemisphere), he let his eyes move around the room. (Both hemispheres have control of the eye muscles.) When the right hemisphere saw something with the same shape as the object it had seen on the screen, it would stop moving the eyes. The left hemisphere just waited for the eyes to stop moving and then called out the shape of the object it saw.

CONCEPT CHECKS

- 13.** Information coming to the left hemisphere of the brain comes from which part of the retinas?
- 14.** After damage to the corpus callosum, a person can describe some of what he or she sees, but not all. Where must the person see something in order to describe it in words? one eye or the other? one half of the retina? one visual field or the other? (Check your answers on page 101.)

Split-brain surgery is extremely rare. We study such patients not because you are likely to encounter one but because they can teach us something about the organization of the brain: Although we cannot fully explain our experience of a unified consciousness, we do see that it depends on communication across brain areas. If communication between the two hemispheres is lost, then each hemisphere begins to act and experience independently of the other.

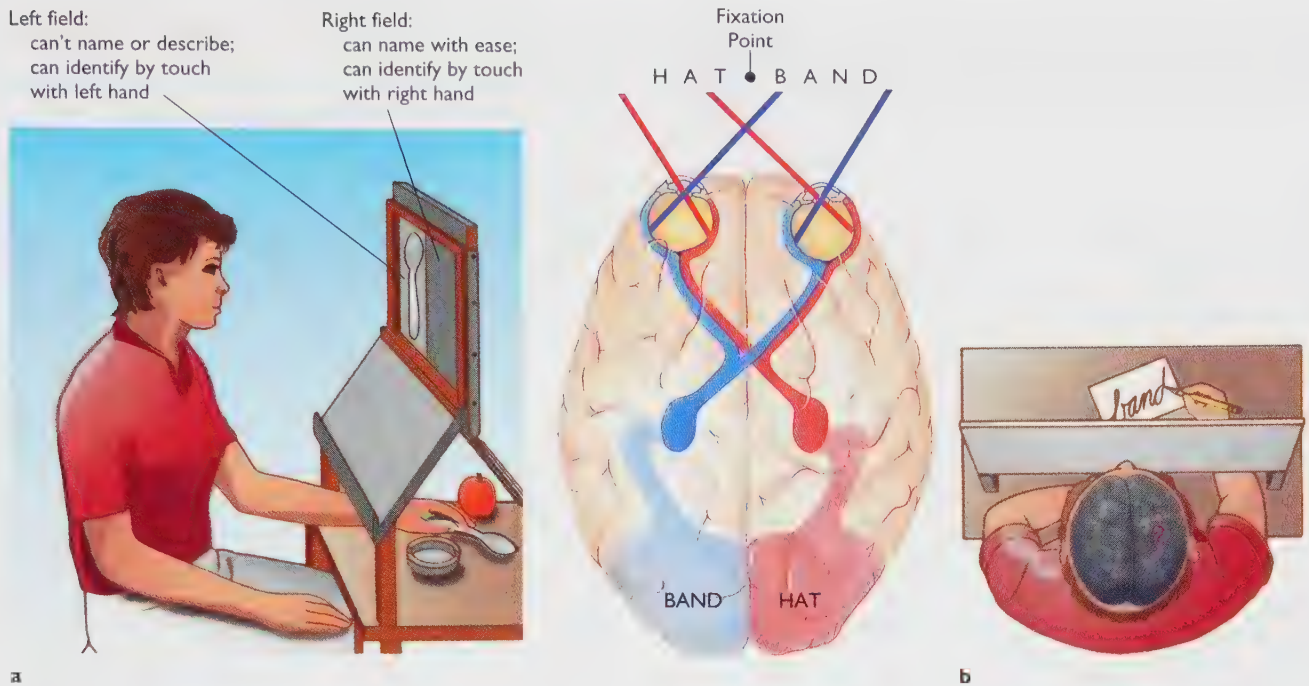


FIGURE 3.43 (a) A woman with a severed corpus callosum cannot name something she sees in her left visual field, but can find the corresponding object with her right hand. (b) When the word *hatband* is flashed on a screen, a woman with a split brain can report only what her left hemisphere saw, *band*. However, with her left hand she can point to a hat, which is what the right hemisphere saw.

Common Misunderstandings About the Brain

Before closing this chapter, I want to caution you about some widespread misperceptions and misunderstandings concerning the brain. One of these concerns the left and right hemispheres. Research on split-brain people and those who have suffered widespread damage to one hemisphere has demonstrated that the two hemispheres have different specializations. The left hemisphere is specialized for language in nearly all right-handers and in most left-handers. The right hemisphere is important for some complex visual-spatial tasks, such as drawing a picture or imagining what something might look like from a different angle. Some writers have gone beyond this generalization, however, to claim that the left hemisphere is logical and the right hemisphere is creative, so some people are logical because they are left-brained people and others are creative because they are right-brained people. Some have even suggested that we could all become more creative if we could find a way to stimulate the right hemisphere or to suppress the competing left hemisphere.

In fact, every task, even the most logical or the most creative, uses both sides of the brain. Certain tasks activate one hemisphere a bit more than the other, but no task relies

on just one hemisphere. As for the theory that some people are left brained or that others are right brained, we simply have no evidence to support that claim. The idea that a highly creative person must be relying mostly on their right hemisphere is merely a guess based on that person's behavior, not based on any measurement of brain activity. It is an example of *neuromythology*—making guesses about the brain to account for an observation about behavior.

A second misunderstanding: You have probably heard the expression “They say we use only 10% of our brain.” Think about that for a moment: Who are “they”? No brain researcher would say anything of the sort. What does this statement really mean? Does it mean that someone could lose 90% of their brain and still function normally? If so, the statement is false. Does it mean that only 10% of the neurons in the brain are active at any given time? False again. Perhaps it simply means that we could all know more and do more than we know and do now. That is undeniably true, though it has nothing to do with the estimated 10% (or any other number). Suppose that you are not much of an athlete; that doesn't mean that you are using only 10% of your muscles. We all use all of our brain and all of our muscles, even when we do not use them very well. Be wary when anyone says “They say that . . .” if the speaker cannot tell you who “they” are or what evidence they have for the conclusion.

THE MESSAGE

Many Brain Areas, One Brain

Most of us act as a single, unified whole (or at most, two wholes, in the case of some people with a split corpus callosum). This unity emerges despite considerable specialization among brain areas, because each area of the brain communicates with the others. Amazingly, an enormous number of separate areas with separate functions, not guided by a master area, alter their functions, based on what the others are doing, to produce coordinated behavior and a single frame of experience—what you call your “self.”

SUMMARY

- ★ **Central and peripheral nervous systems.** The central nervous system consists of the brain (forebrain, midbrain, and hindbrain) and the spinal cord. The peripheral nervous system consists of nerves that communicate between the central nervous system and the rest of the body. (page 85)
- ★ **Autonomic nervous system and endocrine system.** The autonomic nervous system controls the body's organs, preparing them for emergency activities or for vegetative activities. The endocrine system consists of organs that release hormones into the blood. (page 86)
- ★ **Structures of the central nervous system.** Information passing between the cerebral cortex and the spinal cord goes through several intervening structures such as the midbrain and medulla. All the intervening structures have important functions of their own. (page 88)
- ★ **Parallel paths and specialized functions of brain areas.** Contemporary research indicates that the various brain areas have specialized functions and that localized brain damage can produce specialized defects. Various brain functions occur in parallel, largely independently. Loss of part of the brain leads to partial loss of behavior and experience. (page 90)
- ★ **Lobes of the cerebral cortex.** The four lobes of the cerebral cortex and their primary functions are: occipital lobe, vision; temporal lobe, hearing and some aspects of vision; parietal lobe, body sensations; frontal lobe, movement and preparation for movement. (page 93)
- ★ **Imaging brain activity.** Modern technology enables researchers to develop images showing the structure and activity of various brain areas in living, unanesthetized people. (page 95)
- ★ **Corpus callosum.** The corpus callosum is a set of axons through which the left and right hemispheres of the cortex communicate. If the corpus callosum is damaged, information that reaches one hemisphere cannot be shared with the other. (page 96)
- ★ **Connections from eyes to brain.** In humans, information from the *left* visual field strikes the right half of both retinas, from which it is sent to the *right* hemisphere of the brain. Information from the *right* visual field strikes the left

half of both retinas, from which it is sent to the *left* hemisphere. (page 97)

★ **Split-brain patients.** The left hemisphere is specialized for language in most people, so split-brain people can describe information only if it enters the left hemisphere. Because of the lack of direct communication between the left and right hemispheres in split-brain patients, such people show signs of having separate fields of awareness. (page 97)

★ **Common misunderstandings.** Many people try to explain either logical thinking or creative thinking by attributing it to extra activity in the left or right hemisphere of the brain. That guess is at best a half-truth, unsupported by direct evidence. Another common misunderstanding is the assertion that we use only a small percentage of our brain. In fact, people are using all of their brain, even when they are not using it very well. (page 99)

Suggestion for Further Reading

Klawans, H. L. (1996). *Why Michael couldn't hit*. New York: W. H. Freeman. Informative and entertaining account of how the rise and fall of various sports heroes relates to what we know about the brain.

Terms

- central nervous system** the brain and the spinal cord (page 85)
- peripheral nervous system** the bundles of axons that convey messages between the spinal cord and the rest of the body (page 85)
- somatic nervous system** peripheral nerves that communicate with the skin and muscles (page 85)
- spinal cord** that part of the central nervous system that communicates with sensory neurons and motor neurons below the level of the head (page 86)
- sensory neuron** a neuron that carries information about touch, pain, and other senses from the periphery of the body to the spinal cord (page 86)
- motor neuron** a neuron that transmits impulses from the central nervous system to the muscles or glands (page 86)
- reflex** a rapid, automatic response to a stimulus (page 86)
- autonomic nervous system** system of neurons that controls the internal organs such as the heart (page 86)
- endocrine system** a set of glands that produce hormones and release them into the bloodstream (page 87)
- hormone** a chemical released by glands and conveyed by the blood to other parts of the body, where it alters activity (page 87)
- medulla** a structure that is located in the hindbrain and is an elaboration of the spinal cord; controls many muscles in the head and several life-preserving functions, such as breathing (page 88)
- pons** a structure adjacent to the medulla that receives sensory input from the head and controls many muscles in the head (page 88)
- cerebellum** (Latin for “little brain”) a hindbrain structure that is active in the control of movement, especially for complex, rapid motor skills and behaviors that require precise timing (page 88)

- cerebral cortex** the outer surface of the forebrain (page 90)
- hemisphere** the left or right half of the brain; each hemisphere is responsible for sensation and motor control on the opposite side of the body (page 90)
- faceblindness** impairment of the ability to recognize faces, despite otherwise satisfactory vision (page 91)
- color constancy** the ability to continue recognizing colors even after a change in lighting (page 91)
- motionblindness** impaired ability to detect motion in visual perception, despite otherwise satisfactory vision (page 91)
- neglect** the tendency to ignore stimuli on one side of the body or one side of the world (page 91)
- delayed response** a task in which an individual gets a signal and then must wait before making the necessary response (page 92)
- occipital lobe** the rear portion of each cerebral hemisphere, critical for vision (page 94)
- parietal lobe** a portion of each cerebral hemisphere; the main receiving area for the sense of touch and for the awareness of one's own body (page 94)
- primary somatosensory cortex** a strip in the anterior (forward) part of the parietal lobe that receives most touch sensations and other information about the body (page 94)
- temporal lobe** a portion of each cerebral hemisphere; the main processing area for hearing, complex aspects of vision, and certain aspects of emotional behavior (page 94)
- frontal lobe** a portion of each cerebral hemisphere at the anterior pole, with sections that control movement and certain aspects of memory (page 95)
- primary motor cortex** a strip in the posterior (rear) part of the frontal cortex that controls fine movements, such as hand and finger movements (page 95)
- prefrontal cortex** an area in the anterior portion of the frontal lobes, critical for planning movements and for certain aspects of memory (page 95)
- functional magnetic resonance imaging (fMRI)** a technique that uses magnetic detectors outside the head to measure the amounts of hemoglobin, with and without oxygen, in different parts of the brain and thereby provides an indication of current activity levels in various brain areas (page 95)
- corpus callosum** a large set of axons connecting the left and right hemispheres of the cerebral cortex and thus enabling the two hemispheres to communicate with each other (page 96)
- epilepsy** a condition characterized by abnormal rhythmic activity of brain neurons (page 96)

Answers to Concept Checks

- The storage of fats takes place at many sites throughout the body. A hormone diffuses throughout the body; a neurotransmitter exerts its effects only on the neurons immediately adjacent to where it was released. (page 87)
- Although alcohol impairs activity throughout the brain, one of the first areas to show a substantial effect is the cerebellum. The typical symptoms of alcohol intoxication are also symptoms of impairment to the cerebellum. (page 88)
- (a) right parietal lobe; (b) temporal lobe; (c) primary motor cortex of the left frontal lobe; (d) right occipital lobe; (e) prefrontal cortex. (page 95)
- The left hemisphere receives input from the left half of each retina. (page 98)
- To describe something, a person must see it with the left half of the retina of either eye. The left half of the retina sees the right visual field. (page 98).

Web Resources

A Brief Tour of the Brain

suhep.phy.syr.edu/courses/modules/MM/Biology/biology.html
Scientists at Syracuse University review the structure and functions of the brain and its parts, right down to the neurons; includes illustrations and very readable text.

Human Anatomy On-Line

www.innerbody.com/htm/body.html

Click on the picture of the nervous system or on the images button and select a brain image. Go deeper by clicking on the magnifying glass when your cursor touches a hotspot. Animations show the visual, auditory, and other systems in action.

BrainPoke!

www.wlu.edu/~web/bp/brainpk.html

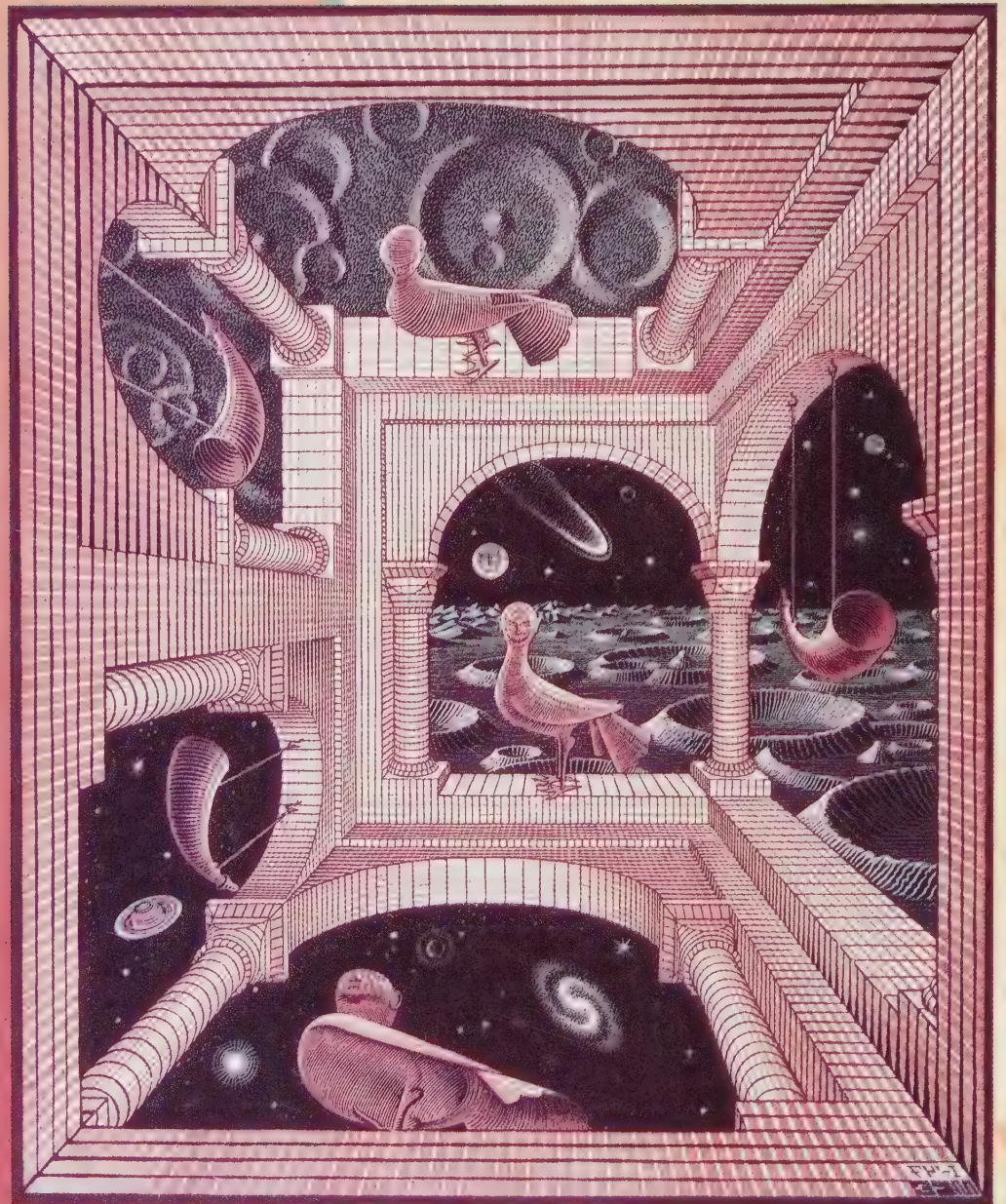
Tell the surgeon where to poke the brain. See the effect. Does the effect match your expectations?

The L.A. Times Brain Page

www.latimes.com/HOME/NEWS/SCIENCE/REPORTS/THEBRAIN/

The *L.A. Times* Brain Page has one of the more impressive pages you will see on the Web. Note the background of the page itself. You will find several interesting brain facts, articles, and links on this page.

Sensation and Perception



4

★ MODULE 4.1

Vision

The Detection of Light

The Structure of the Eye / Some Common Disorders of Vision / The Visual Receptors / Dark Adaptation / The Visual Pathway

Color Vision

The Trichromatic Theory / The Opponent-Process Theory / The Retinex Theory / Colorblindness

Optical Illusions

The Relationship Between Depth Perception and Size Perception / Cross-Cultural Differences in Seeing Optical Illusions / The Moon Illusion

★ MODULE 4.2

The Nonvisual Senses

Hearing

Pitch Perception / Localization of Sounds

The Vestibular Sense

The Cutaneous Senses

Pain

The Chemical Senses: Taste and Smell

Taste / Olfaction

★ MODULE 4.3

The Interpretation of Sensory Information

Perception of Minimal Stimuli

Sensory Thresholds and Signal Detection / Subliminal Perception / What Subliminal Perception Cannot Do / What Subliminal Perception Can Do

Perception and the Recognition of Patterns

*The Feature-Detector Approach / **FEATURE DETECTORS IN THE HUMAN VISUAL SYSTEM: WHAT'S THE EVIDENCE?** Do Feature Detectors Explain Perception? / The Gestalt Psychology Approach / Gestalt Principles in Hearing / Feature Detectors and Gestalt Psychology*

Perception of Movement and Depth

Perception of Movement / Depth Perception

When my son Sam was 8 years old, he asked me, "If we went to some other planet, would we see different colors?" He did not mean just a new shade or a new mixture of familiar colors. He meant colors that were truly new, as different from familiar colors as yellow is from red or blue. I told him that would be impossible, and I tried to explain why.

No matter where we go in outer space, no matter what unfamiliar objects or atmospheres we might encounter, we could never experience a color, or a sound, or any other sensation that would be fundamentally different from what we experience on Earth. Different combinations, perhaps. But fundamentally different sensory experiences, no.

Three years later, Sam told me he was wondering whether people who look at the same thing are all having the same experience: When different people look at some-

thing and call it "green," how can we know whether they are all seeing the same "green"? I agreed that there is no way of knowing for sure.

Why am I certain that colors on a different planet would look the same as they do here on Earth and yet uncertain whether colors look the same to different people here? The answer may be obvious to you. If not, I hope it will be after you have read this chapter.

Sensation is the *conversion of energy from the environment into a pattern of response by the nervous system*. It is the registration of information. **Perception** is *the interpretation of that information*. For example, light rays striking your eyes give rise to sensation. When you conclude from that sensation, "I see my roommate," you are expressing your perception. (In practice, the distinction between sensation and perception is often difficult to make.)

Vision

How do our eyes convert light energy into something that we can experience?

How do we perceive colors?

We live in a world full of **stimuli**—*energies that affect what we do*. Our eyes, ears, and other sensory organs are packed with **receptors**—*specialized cells that convert environmental energies into signals for the nervous system*. We see, hear, and so forth because stimuli activate receptors, which in turn send messages to the brain, which eventually uses this information to guide our behavior.

You have probably already been given this account in a biology or health class in high school, perhaps even in elementary school. But did you believe it? Evidently not everyone does. One survey posed the questions, “When we look at someone or something, does anything such as rays, waves, or energy go out of our eyes? into our eyes?” Among first graders (about age 6), 49% answered (incorrectly) that energy went out of the eyes, and 54% answered that energy came into the eyes. (It was possible to say *yes* to both.) Among college students, 33% said that energy went out of the eyes; 88%, that energy came in (Winer & Cottrell, 1996).

The idea that the eyes send out sight rays is hardly the only widespread misconception about vision. We are often led astray because we imagine that what we see is simply a copy of the outside world, and it is not. For example, color is not a property of objects; it is something your brain creates in response to light of different wavelengths. Brightness is not the same thing as the intensity of the light. (Light that is twice as intense does not appear twice as bright.) Our experiences *translate* the stimuli of the outside world into very different representations.

The Detection of Light

What we call *light* is just one part of the electromagnetic spectrum. As Figure 4.1 shows, the **electromagnetic spectrum** is *the continuum of all the frequencies of radiated energy*—from gamma rays and X rays, which have very short wavelengths, through ultraviolet, visible light, and infrared, to radio and TV transmissions, which have very long wavelengths.

What makes “visible light” visible? The answer is our receptors, which are equipped to respond to wavelengths from 400 to 700 nm. With different receptors, we might see a different range of wavelengths. Some species—bees, for example—respond to wavelengths shorter than 350 nm, which are invisible to humans.

The Structure of the Eye

When we see an object, light reflected from that object passes through the **pupil**, an *adjustable opening in the eye through which light enters*. The **iris** is the *colored structure on the surface of the eye, surrounding the pupil*. It is the structure we describe when we say someone has brown, green, or blue eyes. When the light is dim, muscles open the pupil to let in more light. When the light is bright, muscles narrow the pupil.

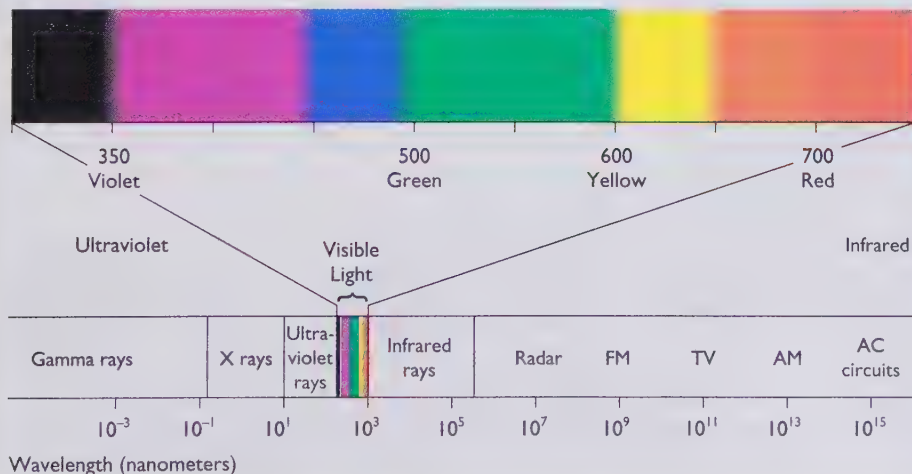
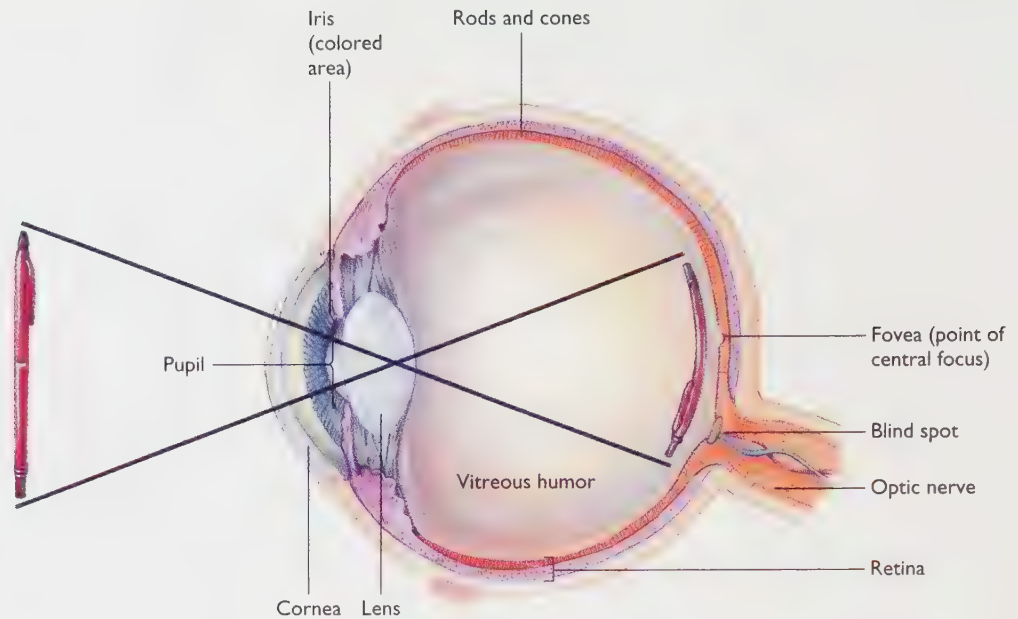


FIGURE 4.1 Visible light, what human eyes can see, is only a small part of the entire electromagnetic spectrum. While experimenting with prisms, Isaac Newton discovered that white light is a mixture of all colors, and color is a property of light. A carrot looks orange because it reflects orange light and absorbs all the other colors.

FIGURE 4.2 The lens gets its name from Latin for *lentil*, referring to its shape—an appropriate choice, as this cross section of the eye shows. The names of other parts of the eye also refer to their appearance.



After light passes through the pupil, it travels through the *vitreous humor* (a clear, jellylike substance) and strikes the retina at the back of the eyeball. The **retina** is a *layer of visual receptors covering the back surface of the eyeball* (Figure 4.2). As light passes through the eye, the cornea and the lens focus the light on the retina as shown.

The **cornea**, a *rigid transparent structure on the outer surface of the eyeball*, always focuses light in the same way. The **lens**, however, is a *flexible structure that can vary in thickness*, enabling the eye to **accommodate**—that is, *to adjust its focus for objects at different distances*. When we look at a distant object, for example, our eye muscles relax and let the lens become thinner and flatter, as shown in Figure 4.3a. When we look at a close object, our eye muscles tighten and make the lens thicker and rounder (Figure 4.3b).

Some Common Disorders of Vision

As people grow older, they gradually develop **presbyopia**, *decreased flexibility of the lens and therefore inability to focus on nearby objects*. (The Greek root *presby* means “old.” This root also shows up in the word *presbyterian*, which means “governed by the elders.”) Many people’s eyes are not quite spherical. A person whose eyeballs are elongated, as shown in Figure 4.4a, can focus well on nearby objects but has difficulty focusing on distant objects. Such a person is said to be **nearsighted** or to have **myopia** (mi-O-pee-ah). About half of all 20-year-olds are nearsighted and must wear glasses or contact lenses to see well at a distance. An older person with both myopia and presbyopia needs bi-focal glasses to help with both near focus and distant focus. A person whose eyeballs are flattened, as shown in Figure 4.4b, has **hyperopia**, or *farsightedness*. Such a person can focus well on distant objects but has difficulty focusing on close objects.

Two other common visual disorders are glaucoma and cataracts. **Glaucoma** is a *condition characterized by increased pressure within the eyeball*; the result can be damage to the optic nerve and therefore a progressive loss of peripheral vision (“tunnel vision”). A **cataract** is a *disorder in which the lens becomes cloudy*. People with severe cataracts can have the lens surgically removed and replaced with a contact lens. Because the normal lens filters out more blue and ultraviolet light than other light, people with artificial lenses sometimes report seeing blue more clearly and distinctly than they ever had before (Davenport & Foley, 1979). They do, however, suffer increased risk of damage to the retina from ultraviolet light.

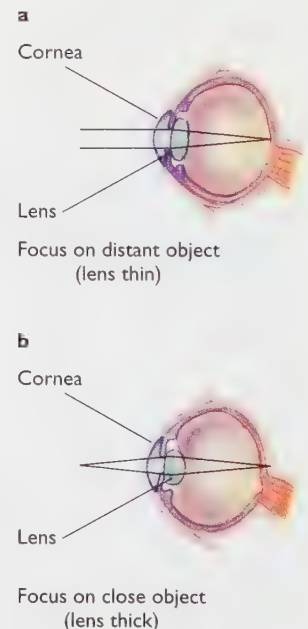
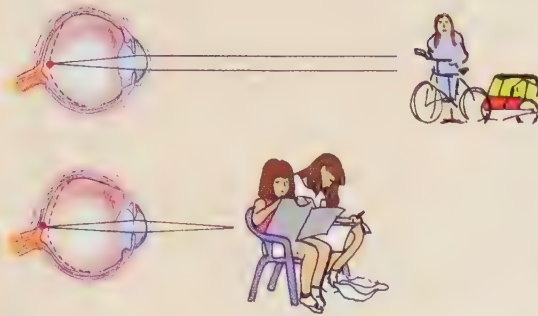


FIGURE 4.3 The flexible, transparent lens changes shape so that objects (a) far and (b) near can come into focus. The lens bends entering light rays so that they fall on the retina. In old age, the lens becomes rigid and people find it harder to focus on nearby objects.

a. Nearsighted (myopic) eyes



b. Farsighted (hyperopic) eyes

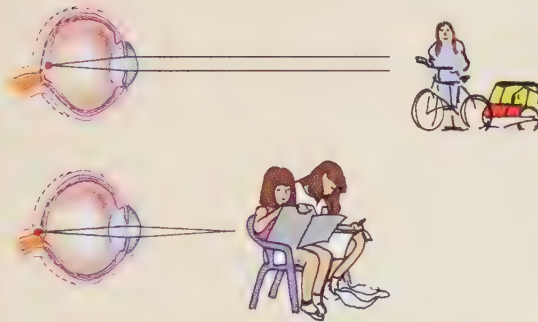


FIGURE 4.4 The structure of (a) nearsighted and (b) farsighted eyes distorts vision. Because the nearsighted eye is elongated, light from a distant object focuses in front of the retina. Because the farsighted eye is flattened, light from a nearby object focuses behind the retina. (The dashed line shows the position of the normal retina in each case.)

CONCEPT CHECK

1. Suppose you have normal vision and you try on a pair of glasses made for a person with myopia. How will the glasses affect your vision? (Check your answer on page 117.)

The Visual Receptors

The visual receptors of the eye, specialized neurons in the retina at the back of the eyeball, are so sensitive to light that they are capable of responding to a single photon, the smallest possible quantity of light. There are two types of visual receptors: cones and rods, which differ in appearance, as Figure 4.5 shows, and in function. The **cones** are *receptors adapted for color vision, daytime vision, and detailed vision*. The **rods** are *receptors adapted for vision in dim light*.

About 5–10% of all the visual receptors in the human retina are cones. Most birds have at least as high a proportion of cones as humans and correspondingly have good color vision. Species that are active mostly at night—rats

and mice, for example—have relatively few cones and a great dominance of rods, thus enhancing their perception in faint light.

The proportion of cones is highest toward the center of the retina. The **fovea** (FOE-vee-uh), *the central area of the human retina*, is adapted for highly detailed vision (see Figure 4.2). Of all retinal areas, the fovea has the greatest density of receptors; also, more of the cerebral cortex is devoted to analyzing input from the fovea than input from other areas. If you want to see something in detail, you focus it on the fovea; for example, you can read letters of the alphabet only if you see them in or near the fovea.

Other animal species have eyes that are organized somewhat differently from human eyes. For example, hawks, owls, and other predatory birds have a greater density of receptors on the top of the retina (for looking down) than on the bottom of the retina (for looking up). When these birds are flying, this arrangement enables them to see the ground beneath them in detail. When they are on the ground, however, they have trouble seeing above themselves (Figure 4.6).



FIGURE 4.5 Rods and cones seen through a scanning electron micrograph. The rods, which number over 120 million in humans, help us see in dim light. The 6 million cones in the retina can distinguish gradations of color in bright light; they enable us to see that roses are red, magenta, ruby, carmine, cherry, vermilion, scarlet, and crimson—not to mention pink, yellow, orange, and white.



FIGURE 4.6 The consequence of having receptors mostly on the top of the retina: Birds of prey, such as these owlets, can see down much more clearly than they can see up. In flight, that arrangement is helpful. On the ground, they have to turn their heads almost upside down in order to see above them.

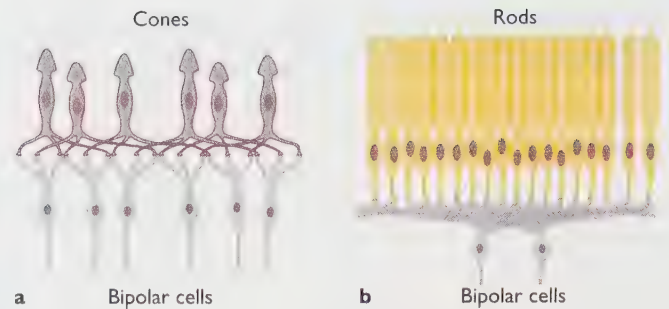


FIGURE 4.7 Only a few cones, but a great many rods, funnel their input into cells of the next layer of the visual system, known as bipolar cells. Because so many rods combine their messages, even a small amount of light falling on the rods can stimulate the bipolar cells. However, the trade-off is that bipolars connected to the cones get relatively precise information about the location of the light, whereas those connected to the rods get only approximate information about the location.

The fovea consists solely of cones (Figure 4.2). Away from the fovea, the proportion of cones drops sharply. For that reason, you have little or no color vision in the far periphery of your eye. Try this experiment: Hold several pens or pencils of different colors behind your back. (Any objects will work as long as they have about the same size and shape and approximately the same brightness.) Pick one at random without looking at it. Hold it behind your head and bring it very slowly into your field of vision. When you just barely begin to see it, you will probably not be able to tell what color it is. (If glaucoma or another medical problem has impaired your peripheral vision, you will have to bring the object closer to your fovea before you can see it at all, and then you will see its color at once.)

The rods are more effective than the cones for detecting dim light, for two reasons: First, a rod is slightly more responsive to faint stimulation than a cone is. Second, the rods pool their resources. In some cases, more than one hundred rods send messages to the next cell in the visual system, whereas only a few cones converge their messages onto a given cell, as shown in Figure 4.7. Table 4.1 summarizes some of the key differences between rods and cones.

CONCEPT CHECK

- Why is it easier to see a faint star in the sky if you look slightly to the side of the star instead of straight at it? (Check your answer on page 117.)

Dark Adaptation

Suppose you go into a basement at night trying to find your flashlight. The only lightbulb in the basement is burned out. A little moonlight comes through the basement windows, but not much. At first you can hardly see anything. A minute or two later, you are beginning to see well enough to find your way around, and eventually you can see well